- The activities that are the responsibility of the blockchain application

Equally significant is the need for blockchain application designers and developers to decide the following:

- Which data will be stored on-chain
- Which data will be stored off-chain

These are the main issues that you'll see addressed in this chapter. In particular, you'll learn about event notification (on-chain data) and how to use it. You should already have an intuitive feel for what is on-chain data. You'll begin by examining different types of on-chain data in the Ethereum blockchain. Next, you'll explore the use of on-chain data in the blind auction Dapp introduced in (chapter 5). Then you'll learn to use both off-chain and on-chain data in the familiar example of the ASK airline Dapp (chapter 2). For each of these applications--blind auction and ASK airline--we'll design and develop an end-to-end application with a web UI.

6.1 On-chain data

Transactions are not the only data stored on the blockchain. A blockchain protocol determines the different types of on-chain data. In Ethereum protocol, as shown in figure 6.2, a block is made up of several elements, each serving a specific purpose:

- The blockchain header (6A) stores the attributes of the block.
- Transactions (6B) store the details of the Txs recorded in the block.